

Article

LightFakeDetect: A Lightweight Model for Deepfake Detection in Videos That Focuses on Facial Regions

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Abstract

In recent years, the proliferation of forged videos, known as deepfakes, has escalated significantly, primarily due to advancements in technologies such as Generative Adversarial Networks (GANs), diffusion models, and Vision Language Models (VLMs). These deepfakes present substantial risks, threatening political stability, facilitating celebrity impersonation, and enabling tampering with evidence. As the sophistication of deepfake technology increases, detecting these manipulated videos becomes increasingly challenging. Most of the existing deepfake detection methods use Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), or Vision Transformers (ViTs), achieving strong accuracy but exhibiting high computational demands. This highlights the need for a lightweight yet effective pipeline for real-time and resource-limited scenarios. This study introduces a lightweight deep learning model for deepfake detection in order to address this emerging threat. The model incorporates three integral components: MobileNet for feature extraction, a Convolutional Block Attention Module (CBAM) for feature enhancement, and a Gated Recurrent Unit (GRU) for temporal analysis. Additionally, a pre-trained Multi-Task Cascaded Convolutional Network (MTCNN) is utilized for face detection and cropping. The model is evaluated using the Deepfake Detection Challenge (DFDC) and Celeb-DF v2 datasets, demonstrating impressive performance, with 98.2% accuracy and a 99.0% F1-score on Celeb-DF v2 and 95.0% accuracy and a 97.2% F1-score on DFDC, achieving a commendable balance between simplicity and effectiveness.

Keywords: deepfake detection; video manipulation; deep learning; multi-task cascaded convolutional network (MTCNN)

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1. Introduction

Deepfake technology employs advanced deep learning methodologies to create highly realistic fake images and videos by altering existing content, resulting in visuals that are indistinguishable to both human observers and automated systems [1]. The term “deepfakes” itself is a combination of “deep learning” and “fakes” [2]. Despite the potential for positive applications in film-making and virtual reality, deepfakes are often misused, particularly in contexts involving high-profile individuals, leading to privacy breaches, harassment, and security risks [3,4]. The first known deepfake video emerged in 2017 when a Reddit user manipulated the faces of public figures for pornographic content [1]. The

rapid increase in deepfakes can be attributed to recent advancements in artificial intelligence technologies (e.g., Generative Adversarial Networks (GANs) [5] and autoencoders [6,7]), which have facilitated the development of sophisticated tools for multimedia manipulation and generation of realistic deepfakes [8].

Various types of deepfakes exist, including face-swapping, which transfers the facial features of one individual to the face of another in a manner that appears seamless and unedited. This technique has been applied across privacy preservation, digital forensics, and entertainment, relying on face alignment, segmentation, and 3D shape estimation to achieve realistic results [9]. Face re-enactment allows the transfer of facial expressions and movements from a source face to a target, maintaining unique identity features while mimicking expressions [10]. Additionally, face reconstruction focuses on generating 3D models from 2D images, with applications in biometrics, film production, and 3D avatars. Historically, high-quality reconstructions required specialized setups, but contemporary methods leverage deep learning and statistical models to achieve this with greater efficiency [11].

In the literature, the problem of deepfake detection has been addressed through a wide spectrum of computer vision and machine learning models, most prominently Convolutional Neural Networks (CNNs) [12,13] for spatial feature extraction, Recurrent Neural Networks (RNNs) [14] for temporal modeling, and more recently, Vision Transformers (ViTs) [15] for capturing both local and global dependencies. Early CNN-based methods, such as Meso-4 and MesoInception-4 [16], targeted mesoscopic facial properties but underperformed on video datasets like Celeb-DF and the Deepfake Detection Challenge (DFDC) due to the absence of temporal modeling [17]. Advances such as the Temporal Dropout 3D CNN (TD-3DCNN) have improved spatio-temporal representation learning through temporal dropout, achieving strong results on the Celeb-DF(v2), DFDC, and FaceForensics++ datasets. Hybrid CNN architectures, including Inception-ResNet v2–Xception combinations [18] and multilayer Xception frameworks [19], have further boosted detection accuracy, though challenges are still posed by low-resolution and compressed videos. RNN-based pipelines, often paired with CNN backbones like EfficientNet [20] or DenseNet [21], have shown the benefits of temporal analysis, while integrations with Transformers and 3D Morphable Models (3DMMs) [22,23] have enhanced sequential modeling capabilities. ViT-based approaches [24–26] have achieved state-of-the-art performance on datasets such as FaceForensics++, Celeb-DF(v2), and WildDeepfake by combining CNN-derived spatial features with transformer-based global context modeling. Complementary to visual analysis, physiological signal-based methods [27–30] leverage eye-blinking patterns and photoplethysmography (PPG) signals to distinguish real from manipulated videos. While these developments demonstrate substantial progress in this field, existing methods often suffer from limited cross-dataset generalization, vulnerability to resolution and compression artifacts, and high computational costs, highlighting the need for lightweight, real-time deepfake detection models.

In this work, we propose LightFakeDetect, a lightweight yet effective deepfake detection framework designed to balance detection accuracy with real-time implementation. Motivated by the strong performance of EfficientNet–GRU-based architectures in capturing spatial–temporal cues [18], we replace the computationally expensive EfficientNet backbone [31] with MobileNet v1 [32], significantly reducing model complexity while preserving its feature extraction capability. We deliberately employ MobileNet v1 as the backbone to prioritize minimal model complexity and the lowest possible number of trainable parameters. Among the MobileNet family, V1 remains one of the lightest and simplest architectures, making it well suited for the objectives of this study. To further enhance spatial feature representation, we integrate the Convolutional Block Attention Module (CBAM) [33], enabling the network to focus on facial regions that can be manipulated.

Temporal dependencies across video frames are modeled using a Gated Recurrent Unit (GRU) [34], which efficiently captures sequential artifacts introduced during manipulation. Our design principle stems from the need for a resource-efficient model that can maintain high detection accuracy without incurring prohibitive computational costs, thereby enabling deployment on edge devices and in real-time scenarios. The novelty of Light-FakeDetect lies in its applied and integrative design. To the best of our knowledge, this is the first work to combine a MobileNet backbone with a CBAM and a GRU specifically for deepfake video detection, while explicitly optimizing the pipeline for class-imbalanced datasets. In addition, we introduce a novel frame preprocessing method that removes duplicate frames or frames with high similarity, rather than relying on random frame removal as in prior work, effectively reducing redundancy while preserving informative temporal content. By demonstrating that this deliberately lightweight architecture achieves performance comparable to or exceeding that of substantially more complex models on large-scale benchmarks, the study provides a practically significant contribution aimed at real-world deployment rather than a purely theoretical advance. The key contributions of this study are (1) the introduction of a MobileNet–CBAM–GRU pipeline tailored for deepfake detection; (2) a carefully designed preprocessing pipeline using a Multi-Task Cascaded Convolutional Network (MTCNN)-based face localization and image pyramids to standardize and enhance frame quality; and (3) a comprehensive evaluation of the model's performance on benchmark datasets (Celeb-DF v2 and DFDC) to demonstrate its effectiveness under realistic constraints.

2. Related Work

Deepfake detection has been approached using a variety of computer vision and machine learning techniques, with Convolutional Neural Networks (CNNs) being the most widely adopted in the literature [12,13]. Early architectures such as Meso-4 and MesoInception-4 [16] were designed to capture mesoscopic facial properties for manipulation detection. While effective for still images, their lack of temporal modeling limited their performance in video-based detection, resulting in relatively low AUC scores of 54.8% on Celeb-DF and 75.3% on DFDC [17]. To address these limitations, the TD-3DCNN method introduced temporal dropout (TD) to selectively omit frames during training, thereby enhancing spatio-temporal feature learning. This approach achieved accuracies of 81.08%, 82.64%, and 79.09% on Celeb-DF(v2), DFDC, and FaceForensics++, respectively. Other CNN-based methods have leveraged more sophisticated backbones. Ikram et al. [18] integrated Inception-ResNet v2 [35] with Xception [36], combining inception modules, residual connections, and depthwise separable convolutions for efficient feature extraction, achieving strong results on DFDC. Similarly, a multilayer Xception-based framework [19] aggregated clues from multiple domains, reaching 92.34% accuracy on Celeb-DF(v2), though its performance remained sensitive to resolution changes. Battula and Rajasekaran [37] proposed a deepfake video detection method using the AdaBoost algorithm, an ensemble learning technique that iteratively combines weak classifiers to focus on difficult-to-classify instances. Evaluated on the DFDC dataset, their approach achieved 86.5% accuracy, 84.7% precision, 88.2% recall, and an F1-score of 86.4%, demonstrating robust performance in detecting subtle manipulations. The study highlights AdaBoost as an effective alternative to CNN-based methods and suggests future integration with CNN feature extractors to further enhance detection capabilities.

Recurrent Neural Networks (RNNs) [14], which excel in modeling temporal dependencies, have also been integrated into deepfake detection pipelines. Montserrat et al. [20] combined MTCNN-based face detection with EfficientNet feature extraction enhanced by ArcFace loss, followed by a Gated Recurrent Unit (GRU) for temporal analysis. An auto-

matic weighting mechanism addressed variability in face quality, yielding 91.88% accuracy on DFDC. Masi et al. [21] proposed a two-branch recurrent network that fuses color- and frequency-domain features via DenseNet [38] and bidirectional Long Short-Term Memory (LSTM), achieving an AUC of 93.18% on FaceForensics++ and of 73.41% on Celeb-DF. Other approaches have incorporated LSTM and Transformers. For example, Petmezas et al. [22] combined CNNs, LSTM, and Transformers with 3D Morphable Models (3DMMs) in a one-class learning framework trained solely on pristine VoxCeleb2 videos [39]. BiLSTM Multi-Head Self-Attention Network (BMNet) [23] further enhanced sequential analysis by integrating BiLSTM with Multi-Head Self-Attention for facial-landmark-based detection, achieving accuracy of 80.20% on Celeb-DF and 84.72% on DFDC.

The emergence of Vision Transformers (ViTs) [15] has led to significant advancements in the field by jointly capturing local image features and global inter-pixel relationships. Souidy et al. [24] demonstrated that combining CNN-based feature extraction with ViT-based global context modeling, alongside majority voting across facial regions, yielded 97% accuracy on FaceForensics++. Khormali et al. [25] proposed an end-to-end ViT framework incorporating patch extraction, RetinaFace-based landmark detection [40], and multi-stream transformer blocks, achieving accuracies of 99.41%, 99.3%, and 81.35% on FaceForensics++, Celeb-DF(v2), and WildDeepfake, respectively. Architectural refinements such as integrating depthwise separable convolutions and Convolutional Block Attention Modules (CBAMs) [33] have further enhanced ViT performance, achieving 87.17% accuracy on Celeb-DF(v2) [26]. CNN-based attention models, such as the one proposed by Dang et al. [41], have also demonstrated improvements, achieving an AUC of 0.712 on Celeb-DF. Ensemble approaches leveraging convolutional attention networks with Mobile Inverted Residual Bottleneck Convolutions, self-attention, CutMix augmentation, and bagging have achieved high accuracy and AUC scores across FaceForensics++ and Celeb-DF, though their computational demands may hinder their real-time deployment [42].

Beyond visual features, physiological signal analysis has emerged as a complementary strategy for deepfake detection. Jung et al. [27] introduced DeepVision, which analyzes eye-blinking patterns, though it is vulnerable to well-simulated blinks. Ciftci et al. [28] exploited photoplethysmography (PPG) signals from facial regions of interest, achieving 76.15% accuracy on FaceForensics++. Wu et al. [29] developed a multi-scale spatio-temporal PPG map with a Mask-Guided Local Attention module and Temporal Transformer, achieving 99.38% accuracy on FaceForensics++. Similarly, Ni et al. [30] extracted biometric signals from four facial regions rich in blood vessels, combining them with a Visual Geometry Group (VGG16) and attention mechanism to achieve 99.2% accuracy. These challenges underscore the need to develop lightweight yet robust deepfake detection models capable of maintaining high accuracy while operating in real time across diverse and resource-constrained settings.

Lightweight yet high-performing deepfake detection models have recently attracted attention to address the limitations of large transformer- or ensemble-based systems in resource-constrained environments. Guo et al. [43] proposed Tiny Deepfake Detection (TinyDF), a compact framework with only 5.38 M parameters and 0.59 G FLOPs that achieves strong intra- and cross-dataset generalization. TinyDF introduces a Pyramid Atrous Aggregation (PAA) module to efficiently capture local–global feature relationships and a Shuffle Fusion Mixer (SFM) module to enhance multi-level feature interaction, while leveraging a Kolmogorov–Arnold network for richer nonlinear feature representation. This design enables the model to outperform heavier architectures across several benchmarks, attaining 93.84% accuracy and 97.73% AUC on FF++(c23) and 68.58% accuracy with 74.90% AUC on DFDC, highlighting its potential for accurate and efficient forgery detection under limited computational resources. Yasir and Kim [44] proposed a lightweight deepfake

detection approach based on multi-feature fusion, integrating Histogram of Oriented Gradients (HOG), Local Binary Pattern (LBP), and KAZE features within classical machine learning classifiers such as Random Forest, Extreme Gradient Boosting (XGBoost), Extra Trees, and Support Vector Machine (SVM). Their method targets compressed social media videos, achieving 92.12% accuracy on FaceForensics++ and 78% on Celeb-DF(v2) while significantly reducing computational cost compared to deep learning models. The study demonstrates that feature-level fusion of texture-based descriptors enables efficient and scalable detection suitable for resource-constrained environments.

In summary, deepfake detection research has explored a range of model architectures, including CNNs for spatial feature extraction, RNNs and their variants (e.g., LSTM, GRUs, BiLSTM) for temporal sequence modeling, ViTs for capturing both local and global dependencies, and multimodal approaches incorporating physiological signals such as PPG for biometric verification. These methods have been evaluated on widely adopted benchmark datasets, including Celeb-DF, Celeb-DF(v2), DFDC, FaceForensics++, VoxCeleb2, and WildDeepfake, enabling comparative performance assessment across manipulation types and resolutions. Despite notable progress in this field, existing techniques face limitations such as reduced generalization to unseen manipulation methods, performance degradation across varying video resolutions and compression levels, and high computational complexity, which impede their real-time deployment. Transformer-based and ensemble models, while achieving state-of-the-art accuracy, often require substantial resources for training and inference, limiting their applicability in edge devices or low-power environments. These challenges underscore the need to develop lightweight yet robust deepfake detection models capable of maintaining high accuracy while operating in real time across diverse and resource-constrained settings.

3. Materials and Methods

3.1. Overview of LightFakeDetect Architecture

The architecture of LightFakeDetect, depicted in Figure 1, is inspired by the work of [20], which utilized EfficientNet B5 [31] combined with a GRU [34] for deepfake detection. This approach demonstrated EfficientNet's efficacy in extracting spatial features and the GRU's ability to capture temporal dependencies across video frames, enhancing their manipulation detection. To streamline the model, we substitute the heavier EfficientNet B5 with the lightweight MobileNet v1 [31,32]. Additionally, we implement a Convolutional Block Attention Module (CBAM) [33], which integrates effectively with MobileNet, replacing the original attention mechanisms. Details of the enhancements are provided in the following sections.

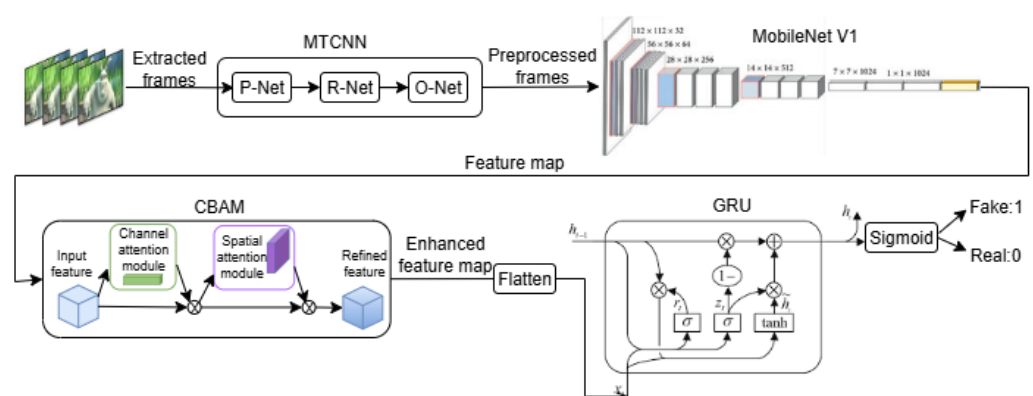


Figure 1. LightFakeDetect architecture. Figure has been reconstructed from Refs. [33,45–47].

As illustrated in Figure 1, the process starts by acquiring video data from established deepfake datasets (Section 3.2). The extracted frames are preprocessed, and an image pyramid is created for each frame (Section 3.3). Using an MTCNN [45], we detect and crop faces, focusing on areas where deepfake manipulations are most evident, thereby enhancing detection accuracy and ensuring consistent input dimensions. Next, we utilize MobileNet [32], a lightweight Convolutional Neural Network, to extract spatial features from the cropped faces (Section 3.1.1). The extracted feature maps are then refined using a CBAM [33]. The CBAM applies both channel and spatial attention mechanisms sequentially, emphasizing the most critical features and enhancing the model's ability to discern subtle manipulations (Section 3.1.2). The next stage involves capturing temporal dependencies using a GRU [34] (Section 3.1.3). Flattened and refined feature maps are fed into the GRU, which models the sequence of frames and captures the temporal relationships that are essential for detecting deepfakes. The last hidden state from the GRU is passed through a fully connected layer, followed by a sigmoid activation function that outputs the final prediction. The key blocks of the architecture are detailed in the following sub-sections.

3.1.1. MobileNet

MobileNet v1 [32], a lightweight and efficient model ideal for fast processing of video data, is used as the feature extractor for each cropped image. Each cropped image, resized to 224×224 , is passed through the first layer of MobileNet.

The MobileNet model leverages depthwise separable convolutions [48], which decompose standard convolutions into two distinct components: depthwise and 1×1 pointwise convolutions. Each input channel in a depthwise convolution is processed with a separate filter, allowing the model to effectively capture spatial features. A pointwise convolution combines the features obtained from the depthwise convolution, facilitating the learning of interactions across different channels. This architecture enables MobileNet to achieve efficient computation and a smaller model size while maintaining strong performance across various tasks.

A depthwise convolution is defined in (1), where G is an input feature map with spatial dimensions of $D_G \times D_G$ and M channels, and $\hat{K}$ is a depthwise convolutional kernel of size $D_K \times D_K \times M$, where the m^{th} channel of the filtered output feature map F is computed by applying the m^{th} filter in K to the m^{th} channel in G .

$$\hat{F}_{k,l,m} = \sum_{i,j} \hat{K}_{i,j,m} \cdot G_{k+i-1,l+j-1,m} \quad (1)$$

MobileNet employs 3×3 depthwise separable convolutions, resulting in a computational efficiency that is 8 to 9 times lower than that of traditional convolutions, with only a minor decrease in accuracy [32]. MobileNet v1 is much smaller and quicker than other models such as VGG16 and GoogleNet [12].

MobileNet comprises 28 layers, beginning with an initial convolution layer followed by several depthwise separable convolution layers. Each convolution is succeeded by Batch Normalization and ReLU activations to enhance training efficiency. The architecture concludes with an average pooling layer, a fully connected layer, and a softmax activation, which are omitted, allowing the model to output feature maps directly.

MobileNet features two key hyperparameters: Width Multiplier (α) and Resolution Multiplier (ρ). α scales the number of channels in each layer from N to αN (where $0 < \alpha \leq 1$), reducing the model's size and computational cost, potentially at the expense of accuracy. α adjusts the input dimensions from $H \times W$ to $\rho H \times \rho W$ (where $0 < \rho \leq 1$), lowering spatial resolution to decrease computational demands, also with possible accuracy trade-offs. These parameters enable a balance between efficiency and performance.

Input frames of size $224 \times 224 \times 3$ are processed by the first layer, which employs a 3×3 convolutional layer with a stride of 2. This operation reduces the spatial resolution from 224×224 to 112×112 while increasing the depth of the feature map to 32. In successive layers, the output dimensions progress to 56×56 , then 28×28 , and 14×14 . The final output from MobileNet when the hyperparameters (α and ρ) are set to 1 is a feature map of size $7 \times 7 \times 1024$ that represents the spatial characteristics of the frame, which is then passed through a Convolutional Block Attention Module (CBAM) to enhance the features.

3.1.2. Convolutional Block Attention Module (CBAM)

The Convolutional Block Attention Module (CBAM), proposed by Woo et al. [49], sequentially applies both channel and spatial attention. Channel attention captures global information using both average pooling (AP) and max pooling (MP) to create two spatial-context descriptors, F_{avg}^c and F_{max}^c . These descriptors are fed into a shared Multi-Layer Perceptron (MLP), and the resulting feature vectors are multiplied element-wise, emphasizing important channels. The channel attention is computed as in (2).

$$M_c(F) = \sigma(MLP(AvgPool(F)) + MLP(MaxPool(F))) \quad (2)$$

Spatial attention generates a spatial attention map by applying AP and MP across the channel dimension, producing two feature maps, F_{avg}^c and F_{max}^c . These maps are concatenated and passed through a convolutional layer to create the spatial attention map (3).

$$M_s(F) = \sigma(f^{7 \times 7}([AvgPool(F); MaxPool(F)])) \quad (3)$$

Here, σ denotes a sigmoid function, and $f^{7 \times 7}$ represents a convolution operation with a 7×7 filter size [33]. After applying both attention mechanisms, a refined feature map is obtained, highlighting the most critical spatial and channel-wise information.

We selected a CBAM as the attention mechanism after evaluating alternatives like Residual Attention Networks (RANs) [50] and Squeeze and Excitation (SE) [51]. The CBAM proved to be lighter and less computationally intensive than the RAN while delivering satisfactory results [33]. Compared to SE, the CBAM excelled by integrating both channel and spatial attention, whereas SE focused solely on channel attention, limiting its ability to guide the model in identifying relevant areas of the feature map for forgery detection.

The CBAM processes the feature map generated by MobileNet. Despite MobileNet using depthwise separable convolutions and a pointwise convolution layer, the output feature map retains sufficient channel information [32] for effective refinement by the CBAM.

3.1.3. Gated Recurrent Unit (GRU)

The output from the CBAM is flattened and passed to a Gated Recurrent Unit [34], a type of RNN employed to process sequence data, making it particularly well-suited for video analysis tasks. We opted for a GRU over LSTM due to its simpler structure with only two gates—the update gate and the reset gate—accelerating the training process while still effectively capturing temporal dependencies [52]. The model signifies how much of the previous hidden state should be used to compute the new candidate hidden state using the reset gate (4), while the update gate (5) decides how much of the hidden state from the previous time step should carry on to the next time step. Equations (4)–(7) describe the calculation performed on the GRU network [52], where r_t , z_t , $\tilde{h}_t$, and h_t represent the reset gate, update gate, candidate hidden state, and current hidden state, respectively; V and W denote the weight matrices; x_t is the input data at time t ; h is the hidden state; and b is the bias.

$$r_t = \sigma(V_r x_t + W_r h_{t-1} + b_r) \quad (4)$$

$$z_t = \sigma(V_z x_t + W_z h_{t-1} + b_z) \quad (5)$$

$$\tilde{h}_t = \tanh(V_h x_t + r_t \oplus (W_h h_{t-1}) + b_h) \quad (6)$$

$$h_t = (1 - z_t) \oplus h_{t-1} + z_t \oplus (\tilde{h}_t) \quad (7)$$

3.2. Datasets

We use two datasets in this study: DFDC [53] and Celeb-DF version 2 [54]. The Celeb-DF v2 dataset comprises three main folders: Celeb-synthesis, which contains 5639 videos; Celeb-real, with 590 videos; and YouTube-real, containing 300 videos. The dataset is divided into two categories: fake and real. The average video duration across the dataset is 12.64 s.

The Deepfake Detection Challenge (DFDC) dataset, created by Facebook AI, comprises 133,154 diverse videos to enhance deepfake detection research. It includes 107,000 deepfake videos (80.36%) and 26,154 real videos (19.64%), divided into training, validation, and test sets. The training set features 119,154 videos, with 100,000 deepfakes generated using techniques like Deepfake AutoEncoder, Morphable Mask, Neural Talking Heads, and faceSwap GAN, alongside 19,154 real videos. The validation set contains 4000 videos, evenly split between the real and fake categories, introducing StyleGAN. The test set includes 10,000 videos, which are also balanced in terms of the real and fake categories. Each video is approximately 10 s long, with an average of 297.56 frames. The distribution of video samples for training and testing the LightFakeDetect model is as follows: For the Celeb-DF (V2) dataset, the training set consists of 5222 videos, including 711 real and 4511 fake videos, while the testing set comprises 1306 videos, with 178 real and 1128 fake videos. In the DFDC dataset, a subset of 5372 videos was selected, with the training set containing 4297 videos (531 real and 3766 fake) and the testing set including 1073 videos (133 real and 942 fake). Additionally, 20% of the training set from both datasets was allocated for validation. Figure 2 presents the distribution of real and fake samples across the Celeb-DF V2 and the DFDC dataset splits.

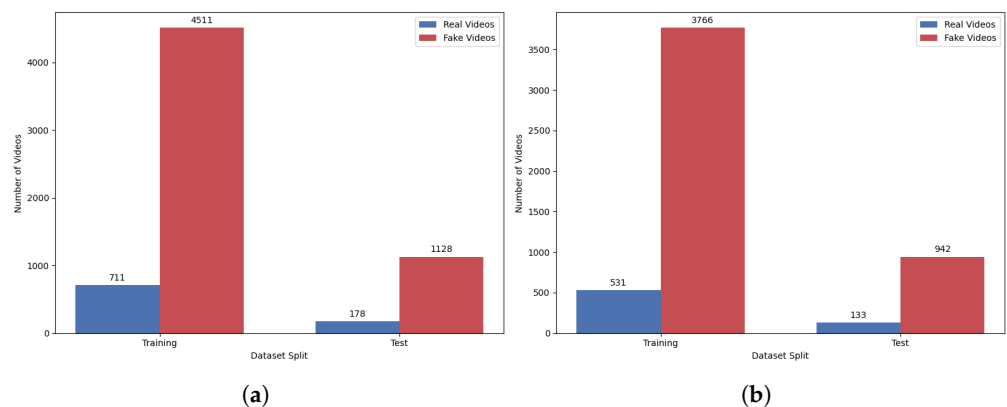


Figure 2. Dataset distribution used in model training and testing: (a) Celeb-DF V2, (b) DFDC.

3.3. Preprocessing Pipeline

During preprocessing of the Celeb-DF v2 dataset, facial regions are detected and cropped using the MTCNN, then resized to 224×224 pixels. Redundant frames are filtered based on the structural similarity index (SSIM) [49], with an initial threshold of 0.85 to remove duplicates while preserving diverse frames. The threshold is adjusted to maintain 40 to 55 frames per video [26], duplicating frames as needed. For videos with fewer than 40 frames, higher SSIM thresholds (up to 0.97) are applied iteratively to retain unique frames. Excess frames are randomly selected to ensure uniformity across the dataset.

For the DFDC dataset, a two-stage processing approach is employed based on the number of video frames. In the first stage, videos with fewer than 100 frames are sent directly to the Multi-Task Cascaded Convolutional Network (MTCNN) [45] for face detection. If more than 55 valid frames are detected, 55 are randomly selected. For videos with over 100 frames, 100 frames are randomly chosen before processing. Approximately 10% of videos yield fewer than 40 usable frames and are reprocessed with the MTCNN on all frames. If needed, the face detection confidence threshold is lowered from 0.99 to 0.90 to ensure sufficient facial feature extraction. Ultimately, each video contains between 40 and 55 frames to achieve consistent input lengths during training.

The MTCNN is an efficient deep learning method for face detection and alignment that utilizes a three-stage cascaded framework [45]. In the first stage, the Proposal Network (P-Net) generates candidate face windows and refines them through bounding box regression and non-maximum suppression (NMS) to eliminate redundancies. P-Net is flexible, requiring no specific input size. In the second stage, the Refine Network (R-Net) filters false positives and further refines bounding boxes with additional regression and NMS, necessitating an input size of 24×24 pixels. In the third stage, the Output Network (O-Net) enhances face alignment by predicting five facial landmarks and requires a 48×48 pixel input. It uses outputs from earlier stages to create patches from normalized images, ensuring accurate localization.

The MTCNN processes input frames of varying sizes using an image pyramid approach, beginning with frames as small as 12×12 pixels. Bounding boxes from the last stage are used to crop images, while landmarks ensure precise alignment. Aligned images are then resized for input into MobileNet v1, optimizing feature extraction in subsequent processing.

3.4. Experimental Protocols

We performed preprocessing of the Celeb-DF v2 dataset using Jupyter Notebook with Python 3.12.4, which offers various libraries for efficient data processing. For the DFDC dataset and model execution, we utilized Google Colab Pro+, which benefits from a high RAM and access to a powerful A100 GPU, ideal for deep learning tasks. The libraries employed included TensorFlow [55] and Keras [56] for building and training the deep learning model, scikit-learn [57] for data splitting and performance evaluation, and joblib [58] for saving and loading model components. The process of choosing the hyperparameters of the model is explained in Section 3.4, and the evaluation strategy of our deepfake detection model is described in Section 3.5.

In the hyperparameter selection process, α and ρ for MobileNet V1 were configured to 1. This selection preserves the standard configuration of MobileNet V1, allowing the model to operate at its full width and resolution.

For the CBAM component, we focused on tuning the reduction ratio, a hyperparameter that regulates the degree of dimensionality reduction in the channel attention module. We tested values of 1, 4, 8, and 16, with 16 serving as the default. Additionally, we set the kernel size of the convolutional layer—which influences the receptive field and spatial resolution of the attention map [33]—to 7, as this is the standard value.

The GRU architecture incorporates several key hyperparameters that significantly affect its ability to process and learn from sequential data. One such hyperparameter is the number of layers, which defines the model's depth and enables it to capture more complex temporal patterns. We evaluated configurations with 1, 2, 3, and 4 layers. Additionally, the number of units in each GRU layer determines the capacity for learning and information storage; we explored values of 128, 256, and 512 for this hyperparameter.

Additionally, we tuned two hyperparameters: learning rate and batch size. The selected learning rate values were 0.01, 0.001, and 0.0001. For batch size, we chose values of 2, 4, and 8. The number of epochs was set to 10 as in [59], with early stopping implemented to prevent overfitting.

We utilized KerasTuner with a Random Search approach [60], enabling the automated and efficient exploration of various configurations to identify the optimal settings based on validation accuracy. The optimal hyperparameter values identified through this tuning process include a reduction ratio of 1, four layers, 128 units, a learning rate of 0.0001, and a batch size of 4.

3.5. Performance Evaluation

We assessed the effectiveness of our deepfake detection model using several key metrics, including accuracy, precision, recall, and F1-score. Let TP represent the number of true positives, FP the number of false positives, TN the number of true negatives, and FN the number of false negatives. Accuracy (Acc) is a widely utilized metric for assessing algorithm performance in classification problems, and is defined as the ratio of correctly classified instances to the total number of observations (8). Balanced accuracy (Acc_B) equally weighs the performance of positive and negative classes (9), and precision (P) reflects the proportion of relevant items among those selected by an algorithm, measuring how many predicted positives are actually correct (10). Recall (R) represents the proportion of relevant data items that are successfully selected (11) and measures how many actual positive observations were correctly identified by the algorithm. The F1-score ($F1$) evaluates an algorithm's performance by integrating both precision and recall. Mathematically, it is defined as the harmonic mean of these two metrics (12). We also incorporated measures of model efficiency, such as the number of parameters, training time, and Floating-Point Operations (FLOPs).

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

$$Acc_B = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \quad (9)$$

$$P = \frac{TP}{TP + FP} \quad (10)$$

$$R = \frac{TP}{TP + FN} \quad (11)$$

$$F1 = 2 \times \frac{P \times R}{P + R} \quad (12)$$

To evaluate the performance of the LightFakeDetect model, we compared it against two benchmark models. The first [18] features an automatic weighting mechanism and shares architectural and preprocessing similarities with our approach. It utilizes an MTCNN for face detection and alignment, followed by a CNN for feature extraction and a GRU for temporal analysis, and was trained on the DFDC dataset. The second model, ViViT [26], was selected for its use of CBAM, making it suitable for comparison. However, ViViT's architecture is more complex due to additional components. This comparison aimed to determine whether our simpler approach could achieve competitive performance. We also compared our results with pretrained models such as Meso-4 [16], Xception [36], and Inception [61].

4. Results

4.1. Assessment of Class Weight Integration with Loss Function

The performance of the proposed LightFakeDetect model was evaluated using two benchmark datasets: Celeb-DF(V2) and DFDC. Both datasets exhibit significant class imbalance, as shown in Section 3.2, characterized by a considerably higher number of fake videos compared to real ones. This imbalance poses a challenge, as it can lead to model bias toward the majority class.

To address this issue, class weights were employed during training, with greater importance assigned to the minority class (real videos) by incorporating higher weights into the loss function. These weights were computed using the `compute_class_weight()` function from the `sklearn` library, ensuring that the model effectively learns from the underrepresented real videos.

The model was trained under two conditions: one without class weights and another with class weights applied. The results, illustrated in Figure 3, highlight the training and validation loss, accuracy, and F1-score over 10 epochs for both configurations. Overall, the model exhibited consistent improvement across all metrics during training. The incorporation of class weights notably enhanced the model's performance on real videos, as evidenced by an increase in accuracy and F1-scores over time, along with a decrease in training and validation loss. The setting with class weights demonstrated superior stability and performance in most instances, particularly in addressing the challenges posed by the imbalanced data.

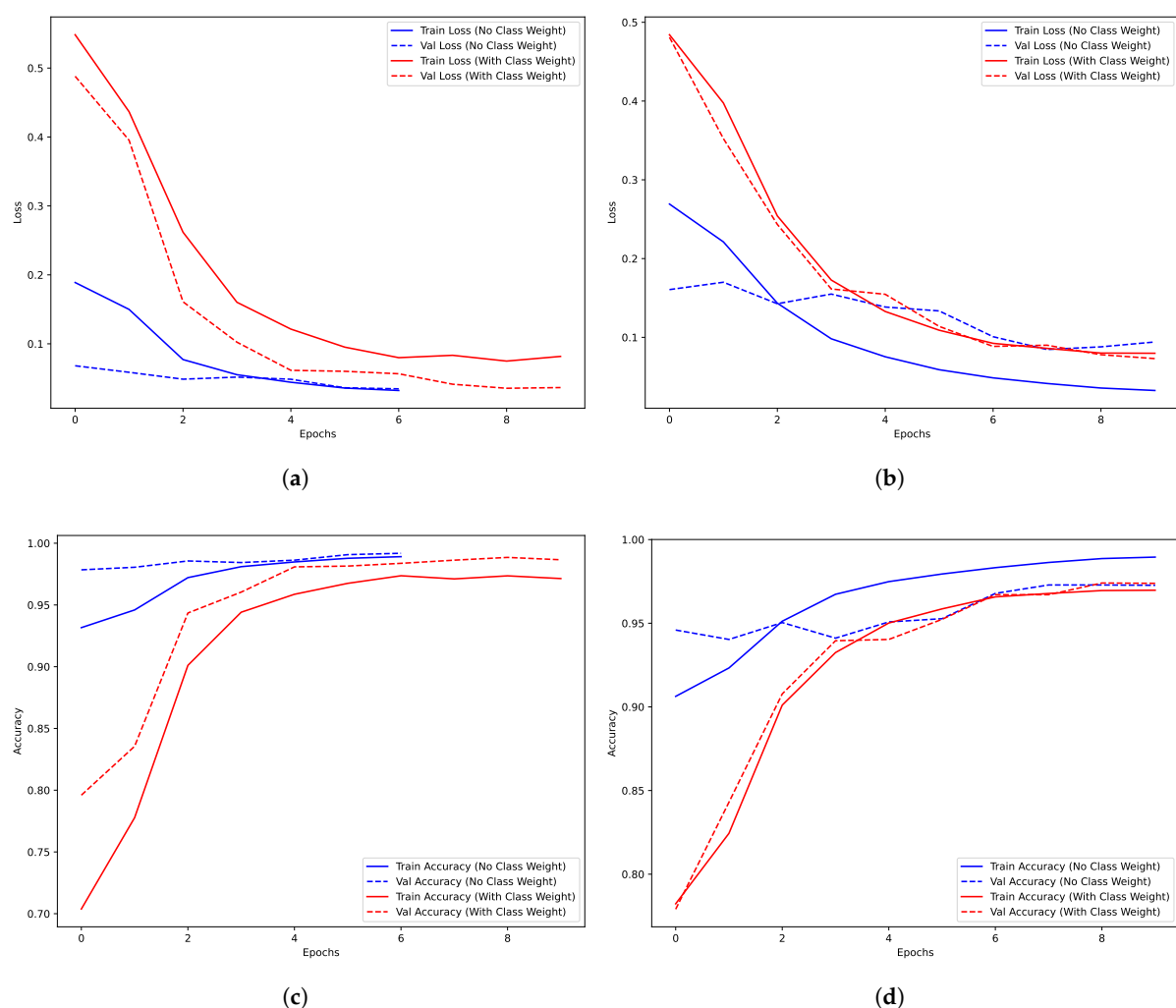


Figure 3. Cont.

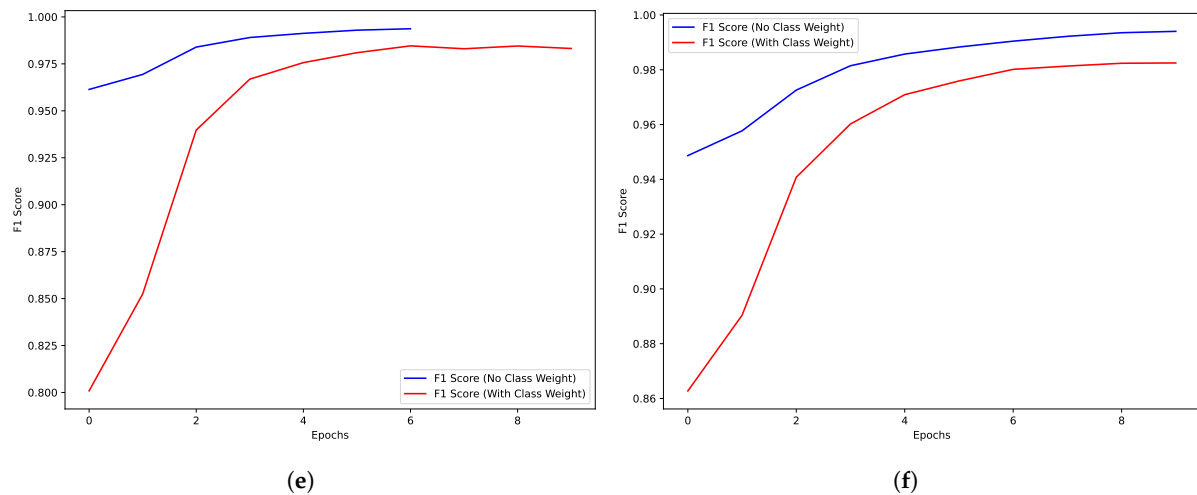


Figure 3. Plots for evaluation of class imbalance mitigation in the LightFakeDetect model: (a) Training and validation loss (Celeb-DF v2). (b) Training and validation loss (DFDC). (c) Training and validation accuracy (Celeb-DF v2). (d) Training and validation accuracy (DFDC). (e) Validation F1-score (Celeb-DF v2). (f) Validation F1-score (DFDC).

The evaluation of the model was conducted on the test sets of the Celeb-DF(V2) and DFDC datasets. Figure 4 presents the confusion matrices, showing how well the model distinguished between real and fake videos under both training settings. The class-weighted model reduced the misclassifications of real videos, particularly on the DFDC dataset.

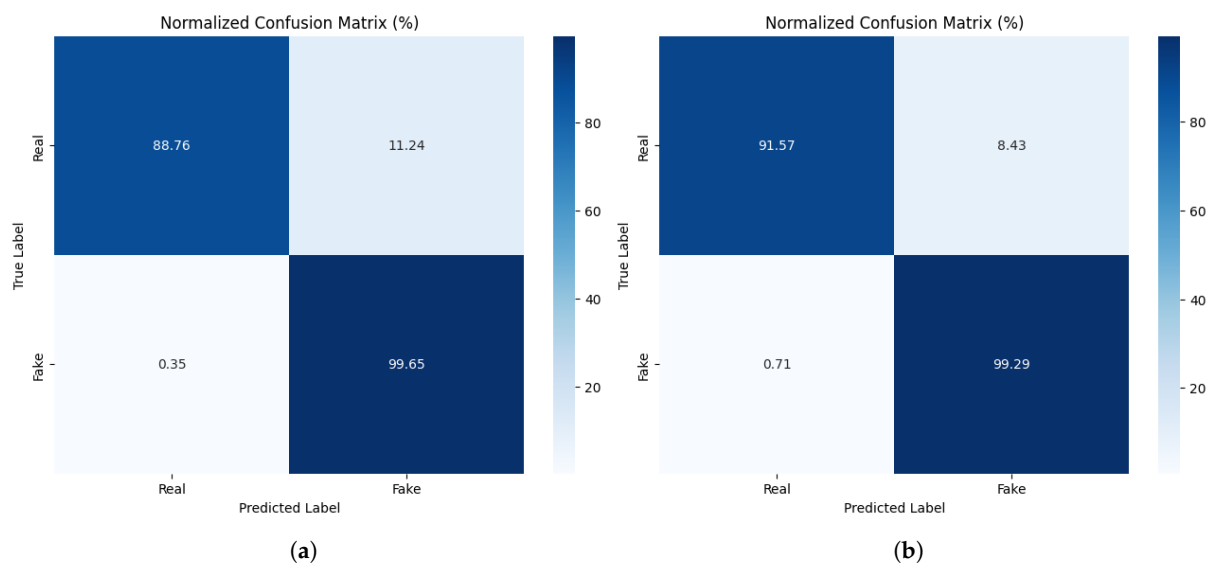


Figure 4. Cont.

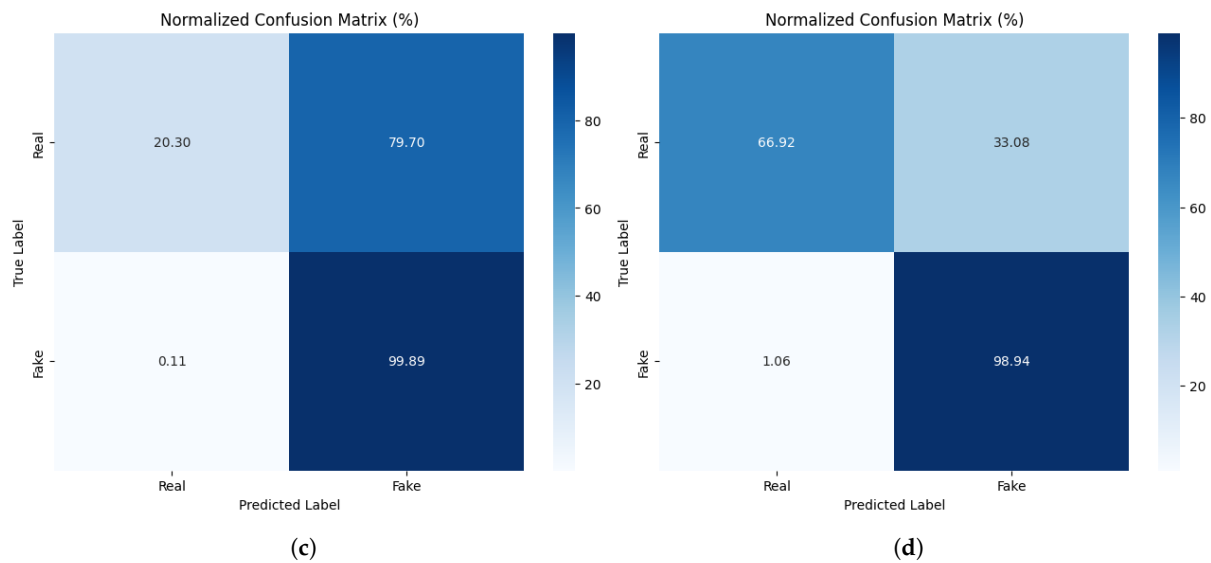


Figure 4. Confusion matrix: (a) Training without using class weights (Celeb-DF v2). (b) Training using class weights (Celeb-DF v2). (c) Training without using class weights (DFDC). (d) Training without using class weights (DFDC).

Figure 5 displays the ROC curves for both datasets. The class-weighted models achieved higher AUC scores, especially on DFDC, indicating better discrimination between classes.

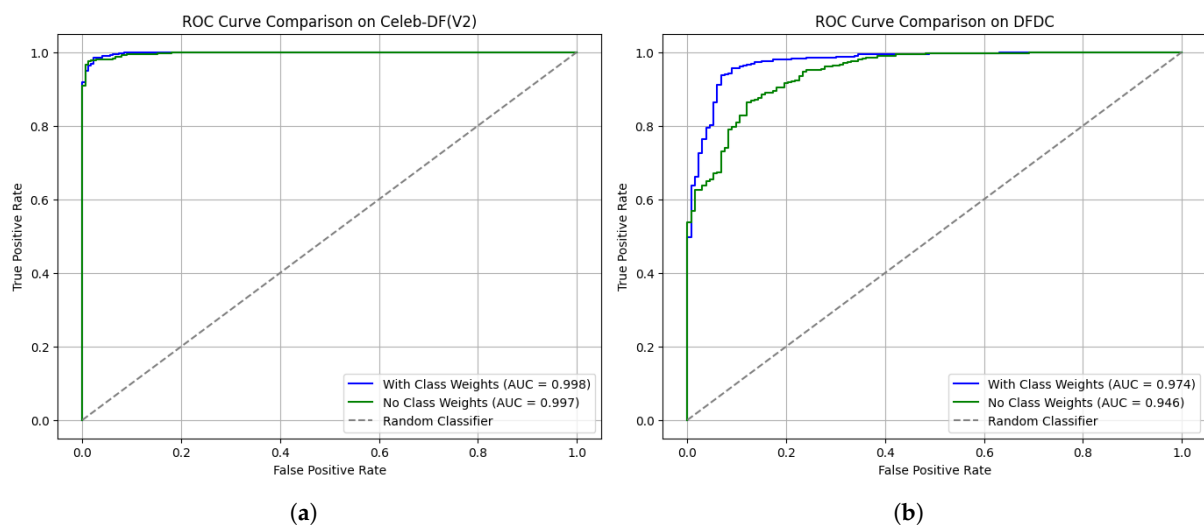


Figure 5. ROC curves for the LightFakeDetect model evaluated on the test sets of the (a) Celeb-DF v2 and (b) DFDC datasets.

Table 1 summarizes the test performance of the DeepFakeDetect model under both training strategies, with and without class weights. On the Celeb-DF(V2) dataset, both approaches achieved high accuracy (Acc) and $F1$ -scores, with the weighted model slightly improving balanced accuracy. For the DFDC dataset, the use of class weights significantly boosted precision (P) and balanced accuracy (Acc_B), indicating better handling of the minority class.

Table 1. Performance evaluation of the LightFakeDetect model on the test set for the Celeb-DF V2 and DFDC datasets, comparing training configurations with and without the application of class weights.

Dataset	Configuration	<i>R</i>	<i>P</i>	<i>F1-Score</i>	<i>Acc</i>	<i>Acc_B</i>	Inference Time	FPS
Celeb-DF	With class weights	99.3%	98.7%	99.0%	98.2%	95.4%	3 s	0.33
	No class weights	99.6%	98.3%	98.9%	98.2%	94.2%	2 s	0.5
DFDC	With class weights	98.9%	95.5%	97.2%	95.0%	82.9%	7 s	0.14
	No class weights	99.9%	89.9%	94.6%	90.0%	60.1%	0.8 s	1.25

All inference times reported in Table 1 correspond to the processing of a single frame (batch size = 1) on an NVIDIA A100 GPU (NVIDIA Corporation, Santa Clara, CA, USA). To facilitate direct comparison with real-time constraints, we converted the average time per frame into frames per second (FPS) using $\text{FPS} = 1/\text{time(s)}$. The resulting FPS values are approximately 0.33 FPS for Celeb-DF with class weights (3 s per frame), 0.5 FPS for Celeb-DF without class weights (2 s per frame), 0.14 FPS for DFDC with class weights (7 s per frame), and 1.25 FPS for DFDC without class weights (0.8 s per frame). These results show that, although LightFakeDetect achieves very high detection accuracy, its current throughput is below typical real-time requirements (around 10–20 FPS). However, because the model is comparatively lightweight, it can be further optimized through quantization, pruning, and GPU-accelerated inference engines (e.g., TensorRT or ONNX Runtime) to reach real-time performance in deployment scenarios.

Although LightFakeDetect contains more parameters than compact CNNs such as Meso-4, it is termed “lightweight” in this study because its design is considerably simpler and less computationally intensive than contemporary transformer-based or multi-stream deepfake detectors. Consequently, while LightFakeDetect has ≈ 22.8 M parameters, it still uses far fewer FLOPs (63.2 GFLOPs vs. 289.2 GFLOPs) and shorter training time (18 h vs. 41 h) than Automatic Face Weighting, making it much more suitable for deployment under constrained computational resources.

4.2. LightFakeDetect Performance Benchmarking

Table 2 presents a comparative analysis of the LightFakeDetect model’s performance against benchmark models, including ViViT [26], Inception [61], Meso-4 [16], Lightweight Deepfake [44], Automatic Face Weighting [20], DeepFake Adaboost [37], and TinyDF [43]. The evaluation utilizes several metrics, such as balanced accuracy (Acc_B), overall accuracy (Acc), area under the receiver operating characteristic curve (AUC), $F1$ -score, Giga Floating-Point Operations (GFLOPs), training time (Time), and the number of parameters (No. Param.).

Regarding the assessment of deepfake detection on the Celeb-DF dataset, LightFakeDetect demonstrates competitive performance in terms of both effectiveness and efficiency. It achieves a balanced accuracy of 95.4% and an $F1$ -score of 99%, which are notably higher than those of traditional models such as Inception and Meso-4, both of which attain accuracies of 86.3% and $F1$ -scores of 92.6%. Additionally, LightFakeDetect has an area under the curve (AUC) of 99.8%, indicating its ability to differentiate between authentic and manipulated content effectively. While ViViT achieves an accuracy of 87.18%, it does not match the robustness or efficiency of LightFakeDetect. The latter also benefits from a more streamlined architecture, as reflected in its lower GFLOPs and processing time. In contrast, the Lightweight Deepfake model falls significantly short, lacking detailed performance metrics but registering only 78% accuracy, which reflects a considerable gap in effectiveness.

Table 2. Performance comparison of benchmark models on the Celeb-DF V2 and DFDC datasets.

Dataset	Model	Acc_B	Acc	AUC	F1-Score	GFLOPs	Time	No. Param.
Celeb-DF	ViViT [26]	-	87.18%	-	92.51%	-	-	-
	Inception [61]	50%	86.3%	58.6%	92.6%	5.694	12 h 5 min	262,401
	Meso-4 [16]	50%	86.3%	55.9%	92.6%	0.094	10 h 5 min	24,137
	Lightweight Deepfake [44]	-	78%	-	-	-	-	-
	LightFakeDetect	95.4%	95.4%	99.8%	99%	63.249	12 h 15 min	22,821,825
DFDC	ViViT [26]	-	88.03%	-	90.23%	-	-	-
	Automatic Face Weighting [20]	74.04%	91.42%	91.80%	95.22%	289.195	41 h 36 min	48,209,658
	Inception [61]	50%	87.6%	62.9%	93.4%	5.694	10 h 15 min	262,401
	Meso-4 [16]	50%	87.9%	63.9%	93.4%	0.094	10 h 38 min	24,137
	DeepFake Adaboost [37]	-	86.5%	-	86.4%	-	-	-
	TinyDF [43]	-	68.58%	74.90%	-	0.59	-	5,380,000
	LightFakeDetect	82.9%	82.9%	97.4%	97.2%	63.249	18 h 28 min	22,821,825

In the DFDC dataset, LightFakeDetect demonstrates competitive performance compared to the other models. It achieves a balanced accuracy of 82.9% and an F1-score of 97.2%, which are respectable results, although it falls short of the accuracy achieved by Automatic Face Weighting at 91.42%. However, LightFakeDetect excels in terms of efficiency, utilizing only 63.249 GFLOPs and requiring 18 h and 28 min for processing. This contrasts sharply with Automatic Face Weighting, which, despite its higher accuracy, demands significantly more computational resources, at 289.195 GFLOPs, and a longer processing time of 41 h and 36 min. Additionally, LightFakeDetect has a more compact architecture with 22,821,825 parameters, compared to the 48,209,658 parameters of Automatic Face Weighting. In comparison, DeepFake Adaboost achieves lower metrics with an accuracy of 86.5% and F1-score of 86.4%. TinyDF, while reporting an accuracy of 68.58% and an AUC of 74.90%, did not train on the DFDC dataset and thus may not reflect its true potential in this context. It does have minimal resource requirements at 0.59 GFLOPs. Overall, while LightFakeDetect may not be the most accurate model, its balance of detection performance and efficiency makes it a compelling choice for practical deepfake detection.

Regarding the comparison of model efficiency, Reference [26] does not provide exact values for the FLOPs or parameter counts of the proposed model. For reference, the standard ViViT-B/16 model contains approximately 89 million parameters and requires about 277.1 to 455.2 Giga Floating-Point Operations (GFLOPs) per video [62]. Since the model presented in this paper is a customized variant of ViViT, its complexity is expected to be comparable to or slightly lower than that of ViViT-B/16. These figures are offered solely for general comparison and should be treated as approximate estimates.

While LightFakeDetect achieves high accuracy on Celeb-DF, its balanced accuracy is lower on DFDC. This discrepancy likely stems from the substantially larger scale and heterogeneity of DFDC, which contains multiple manipulation techniques, varied resolutions, and uncontrolled capture conditions, leading to distributional shifts not present in the training data. Such shifts are known to degrade the performance of deepfake detectors trained on a single dataset. Several strategies can address this limitation. Domain adaptation methods (e.g., adversarial feature alignment or domain-invariant representation learning) could reduce the discrepancy between source and target distributions. Cross-dataset or mixed-dataset training, in which models are simultaneously exposed to multiple sources of manipulated data, can also improve robustness. In addition, augmentation strategies that simulate unseen manipulations, compression artefacts, and noise could further enhance generalization. Investigating these directions is an important avenue for future work to extend LightFakeDetect's applicability to highly diverse or previously unseen forgeries.

LightFakeDetect consistently outperforms existing models across both datasets, offering a strong balance between accuracy and efficiency. These results highlight its suitability for real-world deepfake detection under limited computational resources.

5. Ablation Experiments

To better understand the role of each major component in the proposed LightFakeDetect model, we conducted a series of ablation experiments on the Celeb-DF dataset (see Table 3). In each experiment, we removed or replaced a single module of the architecture while keeping all other training and evaluation settings fixed. This approach allows us to quantify the individual contribution of the CBAM, the GRU temporal module, and the MobileNet backbone to the overall detection performance.

Table 3. Ablation experiments on Celeb-DF.

	Recall	Precision	Accuracy	AUC	Time per Step
Without CBAM	99.47%	99.47%	99.08%	99.9%	10 s
Without GRU	88.91%	99.80%	90.27%	96.3%	6 s
Standard CNN	100%	86.37%	86.37%	47.8%	6 s
Original LightFakeDetect	98.3%	99.6%	95.4%	99.8%	3 s

The full LightFakeDetect model achieved strong results with a recall of 99.6%, precision of 98.3%, accuracy of 95.4% and an AUC of 99.8%. When the CBAM was removed, the model's performance remained essentially unchanged—recall 99.47%, precision 99.47%, accuracy 99.08% and AUC 99.9%, but the training time per step increased noticeably to 10s. This indicates that, under our current settings, although the spatial attention module provides only marginal gains in classification metrics, it contributes to more efficient computation; without it the network becomes significantly slower to train despite producing similar predictive performance.

By contrast, removing the GRU produced a marked degradation in detection ability. The model's recall fell to 88.91%, accuracy to 90.27% and AUC to 96.3% at 6 s per step, although precision slightly increased to 99.8%. These results indicate that the temporal aggregation provided by the GRU is critical for maintaining high sensitivity to manipulated content, especially for videos where frame-to-frame consistency is informative.

The largest performance drop was observed when replacing the MobileNet backbone with a small conventional CNN. Although the model achieved perfect recall of 100%, its precision and accuracy decreased sharply to 86.37% and the AUC collapsed to 47.8% at 6 s per step. This behavior reflects a severe loss of discriminative power and poor calibration of prediction scores, demonstrating that the lightweight MobileNet backbone is essential for producing well-separated feature representations despite its relatively low computational cost.

The ablation results show that although the CBAM module yields only marginal gains in predictive metrics, it substantially improves computational efficiency, as removing it led to a noticeable increase in training time per step despite similar accuracy. In contrast, the GRU module and, in particular, the MobileNet backbone are indispensable for maintaining the high detection accuracy, balanced precision–recall behavior, and robust ranking performance that characterize the full LightFakeDetect model on Celeb-DF.

6. Discussion

The study's final findings demonstrate that the proposed LightFakeDetect model effectively addresses the problem of deepfake video detection, particularly when there is a notable class imbalance. Experiments on two well-known benchmark datasets, DFDC and Celeb-DF(V2), confirm that the model can achieve both high accuracy and stability.

To correct the imbalance between real and fake videos, the model was trained in two different configurations: with and without class weighting. Since better balanced accuracy and precision values were obtained, using class weights proved crucial in enhancing the detection of the minority class (real videos). For instance, using class weights on the DFDC dataset greatly increased the model's precision from 89.9% to 95.5% and its balanced accuracy from 60.1% to 82.9%. This suggests that the model was able to eliminate bias towards the majority fake class and learn more discriminative characteristics for the minority real class.

The strong performance of LightFakeDetect can be attributed to the effective integration of its core components: MobileNet, CBAM, and GRU. MobileNet, a lightweight CNN, efficiently extracts rich spatial features from each video frame, capturing fine-grained facial structures such as textures, edges, and local lighting cues while maintaining low computational overhead. These features are then refined by the CBAM, which sequentially applies channel and spatial attention to highlight the most informative facial regions such as areas exhibiting texture irregularities, unnatural lighting, or blending artifacts, while suppressing less relevant background information. This targeted refinement improves the discriminative quality of the extracted features. Finally, the GRU processes the temporally aligned, refined features across consecutive frames to model motion dynamics and temporal dependencies. This enables the model to detect subtle inconsistencies in facial expressions, lip-syncing, or movements introduced by deepfake manipulations that may not be evident from individual frames. By integrating these three components, LightFakeDetect combines efficient spatial representation, focused attention, and temporal modeling to achieve robust and lightweight deepfake detection.

Accuracy, *F1*-score, and AUC were among the performance metrics on both datasets that continuously improved during training, indicating the model's capability for generalization. LightFakeDetect demonstrated competitive or better accuracy and *F1*-scores compared to baseline models like ViViT and Automatic Face Weighting despite being smaller and requiring less computing power. This is important given the resource constraints that exist in real-world deployment. These findings are supported by the confusion matrices and ROC curves, which show improved class separation and fewer misclassifications under the class-weighted training scenario.

To further analyze the model's limitations, we examined both correct and incorrect classifications in the Celeb-DF dataset. Out of 21 misclassified videos, 19 were real videos predicted as fake, indicating a stronger tendency toward false negatives, due to the lack of real samples in Celeb-DF v2. To better understand these outcomes, we compared representative examples of correct and incorrect predictions. Misclassifications frequently occurred in videos with abrupt cuts, occluded features, or degraded visual quality. A scenario-based inspection revealed several recurring challenges that contributed to misclassification. First, occlusions, shadows, or changes in illumination often distorted facial features, reducing detection reliability; see Figure 6. Second, low-resolution or distant faces led to poor-quality crops by MTCNN, see Figure 7, which limited the detail available for classification. Finally, the unexpected appearance of new faces confused the model's sequence-level analysis.

Figure 6 represents consecutive frames from videos that were wrongly classified as fake, while in fact they are real. This misclassification occurred because of a change in the illumination, which caused the model to interpret the videos as fake. As shown in Figure 7a, the face was captured from a close distance, while in Figure 6b the camera was positioned farther away. The greater distance resulted in low-resolution facial crops, which provided insufficient detail for accurate classification and ultimately led to misclassification.

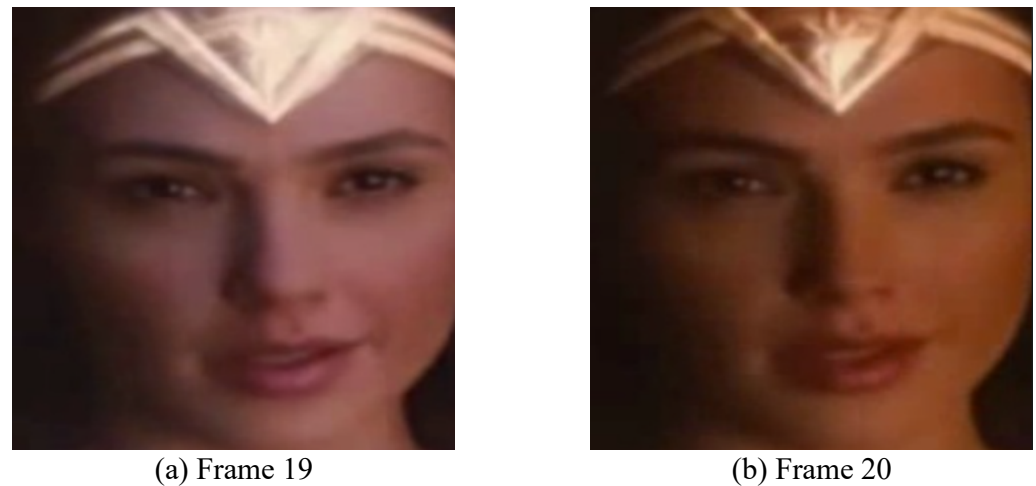


Figure 6. Consecutive frames from videos misclassified as fake due to lighting variations.

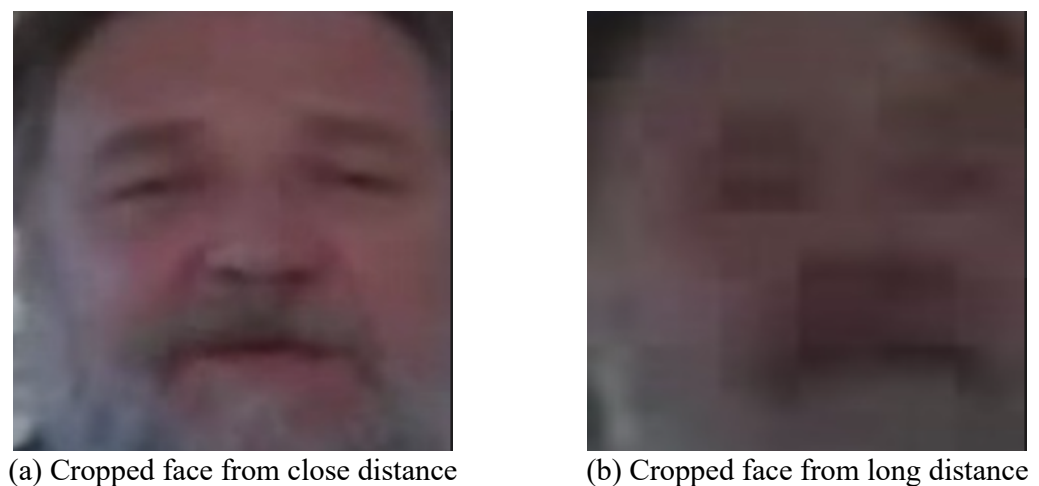


Figure 7. Comparison of facial capture distances: (a) close-range capture providing high-resolution detail, and (b) distant capture resulting in low-resolution facial crops. The decreased detail in (b) contributed to insufficient information for accurate classification, leading to misclassification.

The current results are reported for the full-precision (FP32) LightFakeDetect pipeline. Although our system is not yet implemented as a single end-to-end model, each of its components (MobileNet backbone, CBAM, and GRU) is compatible with widely used model-compression and quantization frameworks (e.g., TensorFlow Lite, ONNX). The literature reports that FP16 quantization can roughly halve model size and increase throughput on embedded GPUs with $<1\text{--}2\%$ loss in F1-score, while INT8 quantization provides even higher speed-ups with slightly larger (2–4%) accuracy drops. We therefore expect LightFakeDetect to retain high detection performance when ported to small devices after such optimizations. A complete evaluation of these trade-offs is planned for future work.

Although LightFakeDetect in its present form analyzes only the visual modality, many deepfake videos also exhibit artifacts in their audio tracks, such as mismatched lip movements, altered prosody, or inconsistent background noise. Combining audio-based features (e.g., spectrogram or Mel-frequency cepstral coefficients extracted from the soundtrack) with our CBAM–GRU video pipeline could capture such inconsistencies and provide complementary cues for discrimination. For instance, an audio branch using a lightweight convolutional network or a transformer encoder could process temporal acoustic patterns in parallel with the video branch, and the outputs could be fused at the feature or decision level. This multimodal approach has been shown in related studies to

boost robustness against sophisticated forgeries and is a promising direction for future work on LightFakeDetect.

The current pipeline relies on an external MTCNN module for face detection and alignment to produce standardized crops for feature extraction. Although MTCNN is fast and accurate for most frontal faces, it can misdetect or fail on occluded, low-quality, or extreme-pose frames, which may propagate errors into later stages of the pipeline. This reliance constitutes a potential limitation of LightFakeDetect's robustness in unconstrained settings. Future work will explore (i) incorporating more resilient detectors (e.g., RetinaFace, BlazeFace) or ensemble strategies to reduce single-detector failure, (ii) adding temporal smoothing to handle intermittent detection errors, and (iii) developing an end-to-end framework that jointly optimizes face localization and deepfake classification to further increase robustness.

Results confirm the initial hypothesis that a lightweight model with balanced training can efficiently detect deepfakes, even when faced with performance variances caused by dataset size and training conditions. The combination of efficient feature extraction, focused attention, and temporal modeling makes LightFakeDetect a reliable, scalable, and high-performing solution for real-world deepfake video detection.

7. Research Implications

The results of this research hold significant implications for both the academic community and real-world applications in the fight against deepfake media. By integrating MobileNet for efficient feature extraction, a CBAM for enhanced spatial feature refinement, and a GRU for temporal sequence modeling, the proposed model achieves a balance between high detection accuracy and computational efficiency. This balance is critical for enabling the model's deployment in resource-constrained environments such as mobile devices, embedded edge devices, and real-time monitoring platforms.

From a practical perspective, the model's ability to achieve accuracy rates exceeding 98% on the Celeb-DF v2 and 95% on the DFDC benchmark datasets demonstrates its robustness against sophisticated and diverse manipulation techniques. This suggests that it has strong potential for integration into social media platforms, digital forensics pipelines, and governmental security infrastructures, where rapid and reliable verification of video authenticity is essential. Furthermore, the use of a lightweight architecture mitigates the operational costs typically associated with large-scale AI deployment, enabling the model's widespread adoption without extensive hardware upgrades.

On a societal level, the proposed framework contributes to countering the growing threats posed by malicious deepfake usage, including political misinformation, reputational damage, and manipulation of legal evidence. By lowering the barrier to deploying effective detection systems, the proposed approach can serve as a scalable defense mechanism, complementing policy measures and public awareness campaigns. The model's design also allows for straightforward retraining and adaptation to emerging manipulation techniques, ensuring that the model's detection capabilities evolve alongside deepfake generation technologies.

8. Conclusions

This paper presents LightFakeDetect, a deep learning model designed to detect deepfakes in video content, addressing the increasing prevalence of manipulated videos. Through a thorough review of existing models and their limitations, we constructed a five-stage pipeline that integrates face detection, feature extraction, attention mechanisms, and temporal analysis. Our model achieved commendable performance metrics, with a balanced accuracy of 95.4% and an *F1*-score of 99% on the Celeb-DF v2 dataset and a balanced accuracy of 82.9% and an

F1-score of 97.2% on the DFDC dataset. LightFakeDetect outperformed benchmark models while maintaining compatibility with devices with limited hardware resources. However, the proposed LightFakeDetect is not a fully end-to-end pipeline, as it relies on a separate face detection and cropping stage using an MTCNN, making it sensitive to errors in preprocessing. Moreover, while MobileNet v1 offers efficiency, its reduced representational capacity may limit detection of subtle or high-quality manipulations, and GRU-based temporal modeling may underexploit complex spatio-temporal patterns. Future research could explore integrating lightweight end-to-end architectures that combine face localization, spatial–temporal modeling, and attention in a single framework. In addition, future research could extend the model to incorporate audio cues for multimodal detection, thus enhancing the robustness of the model against complex forgeries.

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Data Availability Statement: The datasets used in this study are openly available at <https://ai.meta.com/datasets/dfdc/> (accessed on 12 June 2025) and on Github at <https://github.com/yuezunli/celeb-deepfakeforensics?tab=readme-ov-file> (accessed on 15 June 2025).

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